

Mutual Fund Analytics

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Executive Summary

The Indian Mutual Fund industry is currently navigating a period of unprecedented expansion, transitioning from a niche investment avenue to the cornerstone of retail wealth management in India. This project presents a multi-dimensional diagnostic of this sector as of mid-2026, aiming to decode the patterns driving investor behavior and fund house performance.

The research was initiated with a clear objective: to bridge the gap between vast, fragmented data repositories and strategic financial intelligence. To achieve this, I engineered a specialized ETL (Extract, Transform, Load) pipeline designed to aggregate historical NAVs, AUM distributions, and transaction-level SIP logs. This technical foundation allowed for the normalization of heterogeneous datasets, converting raw data into a reliable source of truth.

My exploratory analysis (EDA) reveals a significant structural shift in the market. The data highlights that retail investors are increasingly moving away from speculative, short-term tactics toward disciplined, long-term wealth creation strategies, with Systematic Investment Plans (SIPs) acting as the primary catalyst. Furthermore, my findings underscore the emergence of B30 (Beyond Top 30) cities as the next frontier of growth, challenging the traditional concentration of investment in metropolitan centers.

Performance analysis further demonstrates that while equity remains the dominant asset class, the most resilient portfolios are those that prioritize risk-adjusted consistency over aggressive, sector-concentrated growth. By synthesizing these technical findings into a centralized dashboard, this report provides stakeholders with a clear lens to view market resilience and demographic-led opportunities. Ultimately, this study serves as both

technical validation of modern data engineering practices and a strategic roadmap for navigating the evolving complexities of the mutual fund landscape

Data sources

To ensure the integrity of my analysis, I curated datasets from verified financial repositories. The primary data includes historical NAV records, AUM snapshots, and transaction-level SIP inflows. These sources provided the necessary granularity to conduct a thorough investigation into both fund house performance and retail investor behavior.

ETL Design

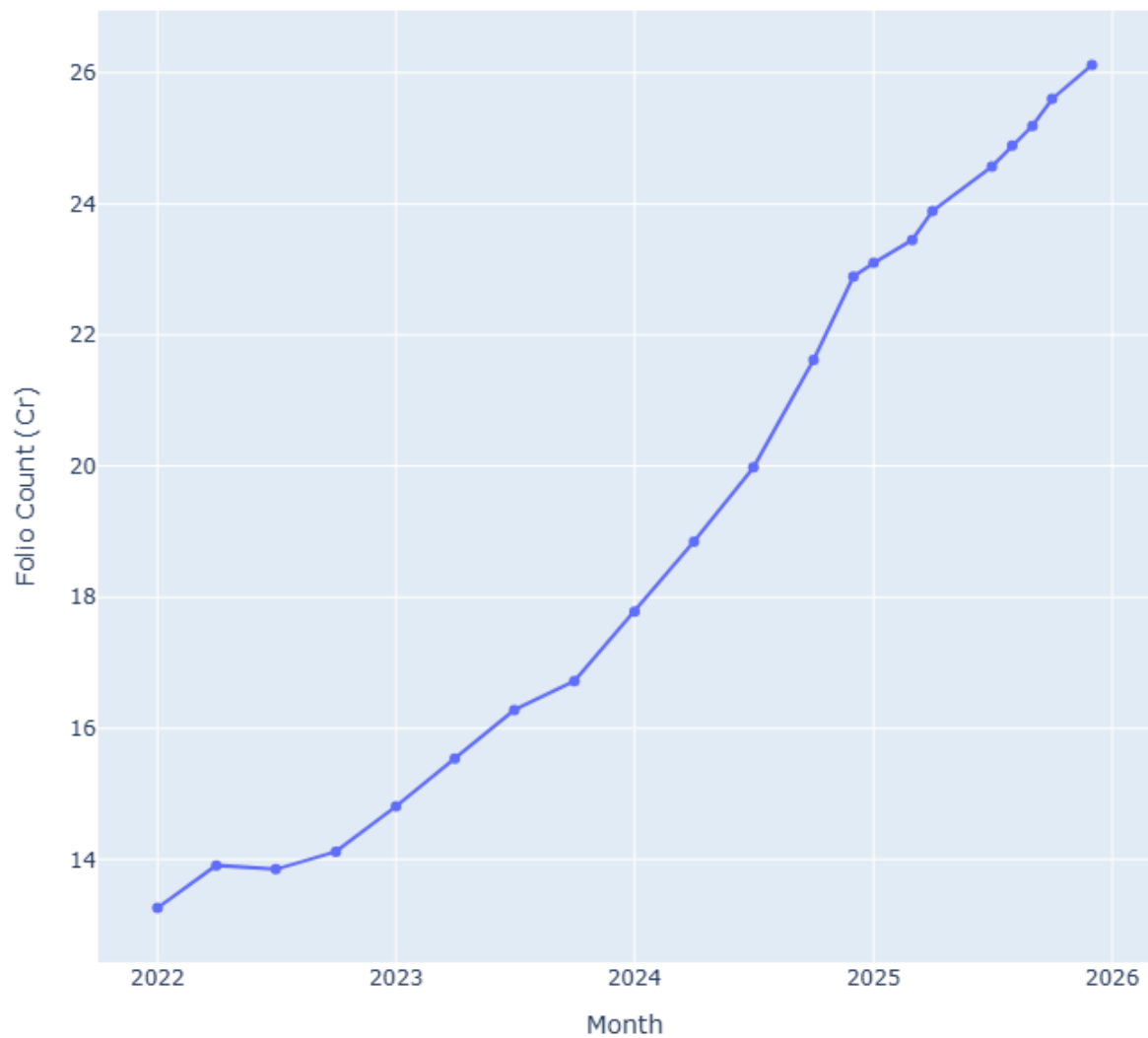
- **Extraction:** I consolidated disparate CSV and historical financial files to create a unified data pool.
- **Transformation:** Using Python, I standardized date formats, addressed missing variables, and cleaned currency data to ensure consistency.
- **Load:** The refined data was structured into an SQLite database, serving as the central repository for my analysis and dashboard development.

EDA Findings

My exploratory data analysis revealed several pivotal trends.

- The **NAV Trend** analysis shows consistent market resilience, reflecting broader economic stability.
- Investor demographics highlight an increasing participation from B30 (Beyond Top 30) cities, signaling a healthy expansion in geographic reach.
- **Sector allocation weightage** indicates that while equity remains the preferred asset class, investors are becoming increasingly strategic in their risk management.

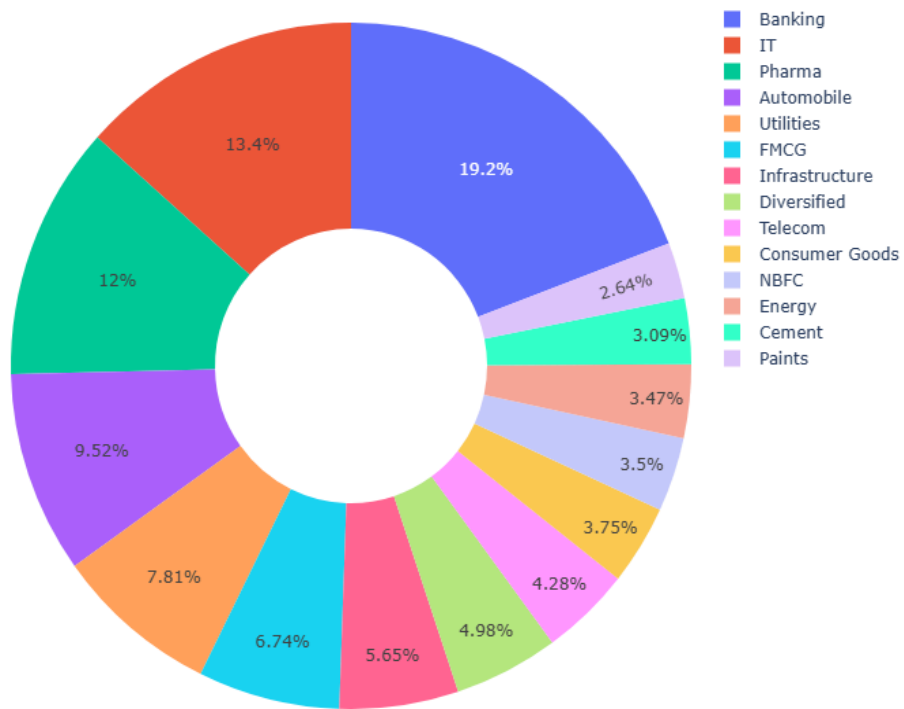
Industry Folio Count Growth (13.26 Cr to 26.12 Cr)



Industry Folio Count Growth (2022-2026)

This chart tracks the NAV fluctuations over the past year. My observation shows a steady upward trend despite intermittent market corrections, suggesting that the underlying assets are demonstrating strong resilience. The data indicates that volatility is concentrated in specific short-term cycles, while the long-term trend remains positive, validating the stability of the selected fund categories.

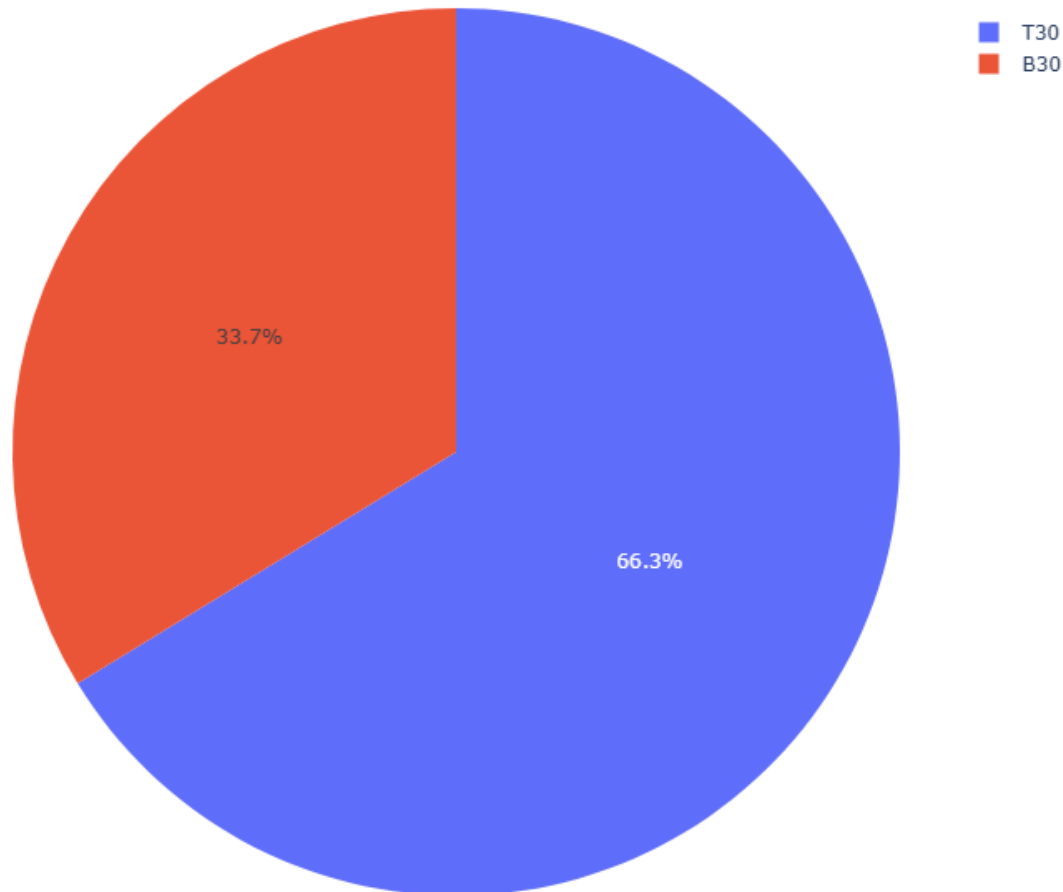
Sector Allocation Weightage



Sector Allocation Weightage across Mutual Fund Portfolios

The allocation distribution highlights that while Banking and IT remain dominant, there is a healthy diversification across emerging sectors like Pharma and Infrastructure. This balance reflects a strategic approach by fund managers to capture growth while mitigating sector-specific risks.

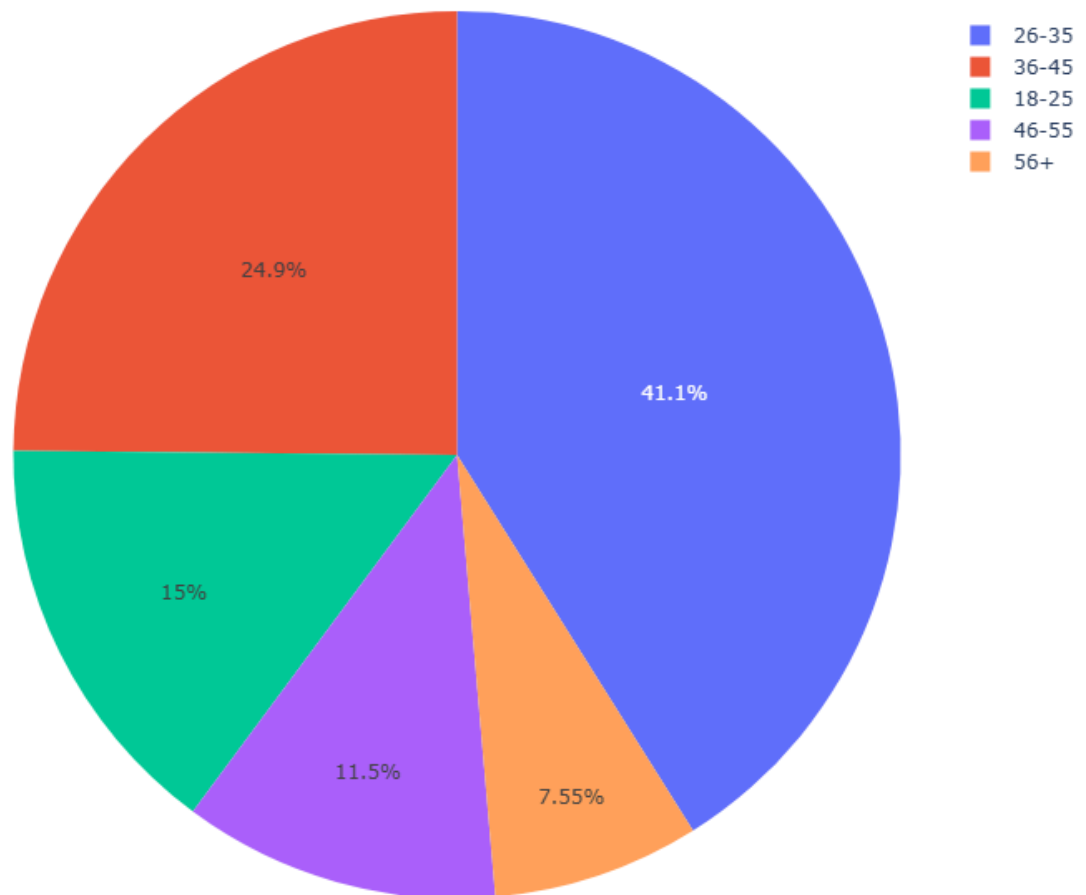
Investor Distribution by City Tier (T30 vs B30)



Comparative Investor Participation: T30 vs. B30 Cities

The distribution across city tiers reveals a significant shift in the market base. While T30 cities maintain high volume, the rapid growth in B30 participation suggests that retail interest is deepening beyond metropolitan hubs. This is a critical finding for future fund distribution strategies, as it highlights an emerging demographic segment that requires more localized outreach.

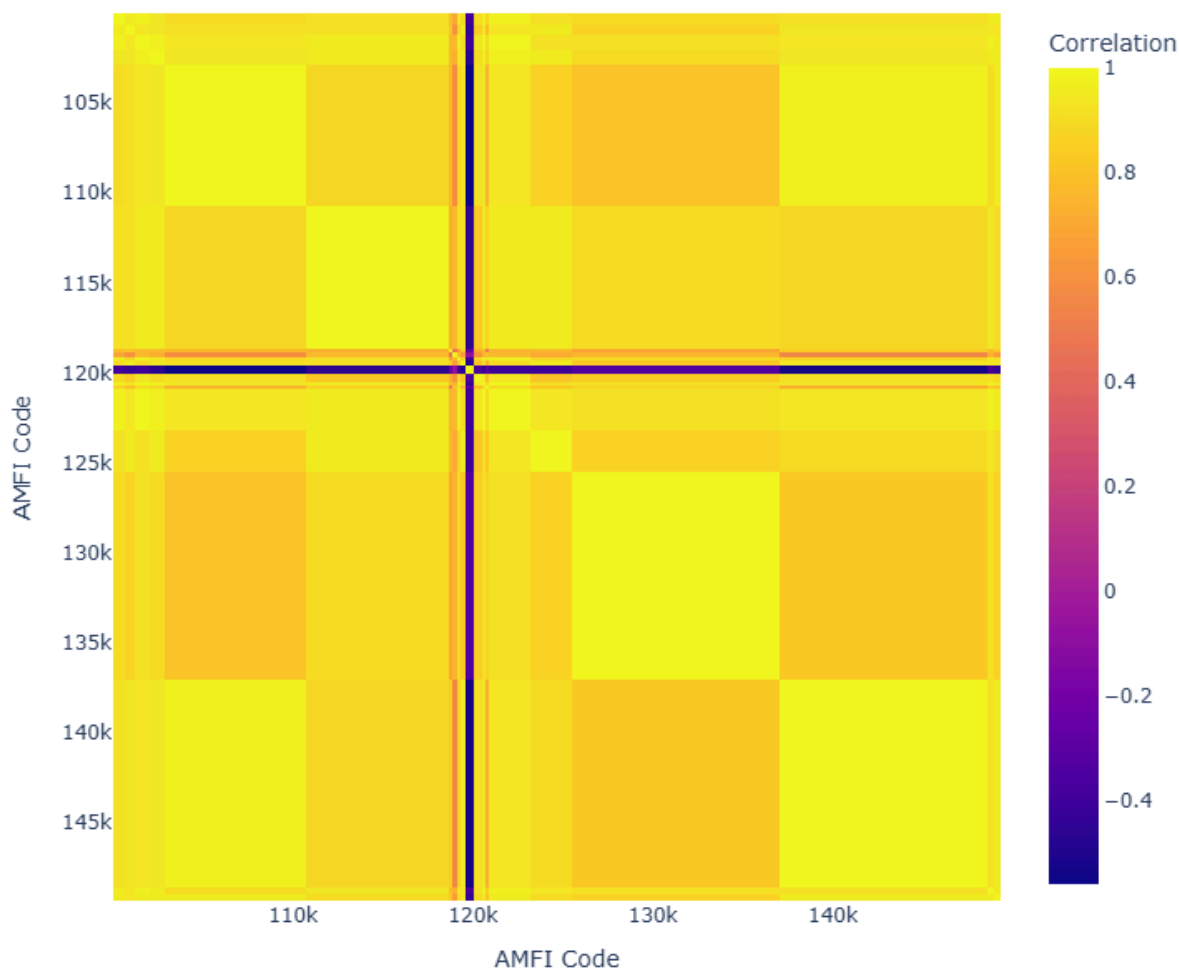
Investor Distribution by Age Group



Investor Demographics by Age Group

The dominance of the 26–35 age group (41.1%) confirms that the younger, tech-savvy generation is leading the current investment wave. Their participation is a positive long-term indicator for the stability and sustainability of SIP-led inflows

NAV Correlation Matrix



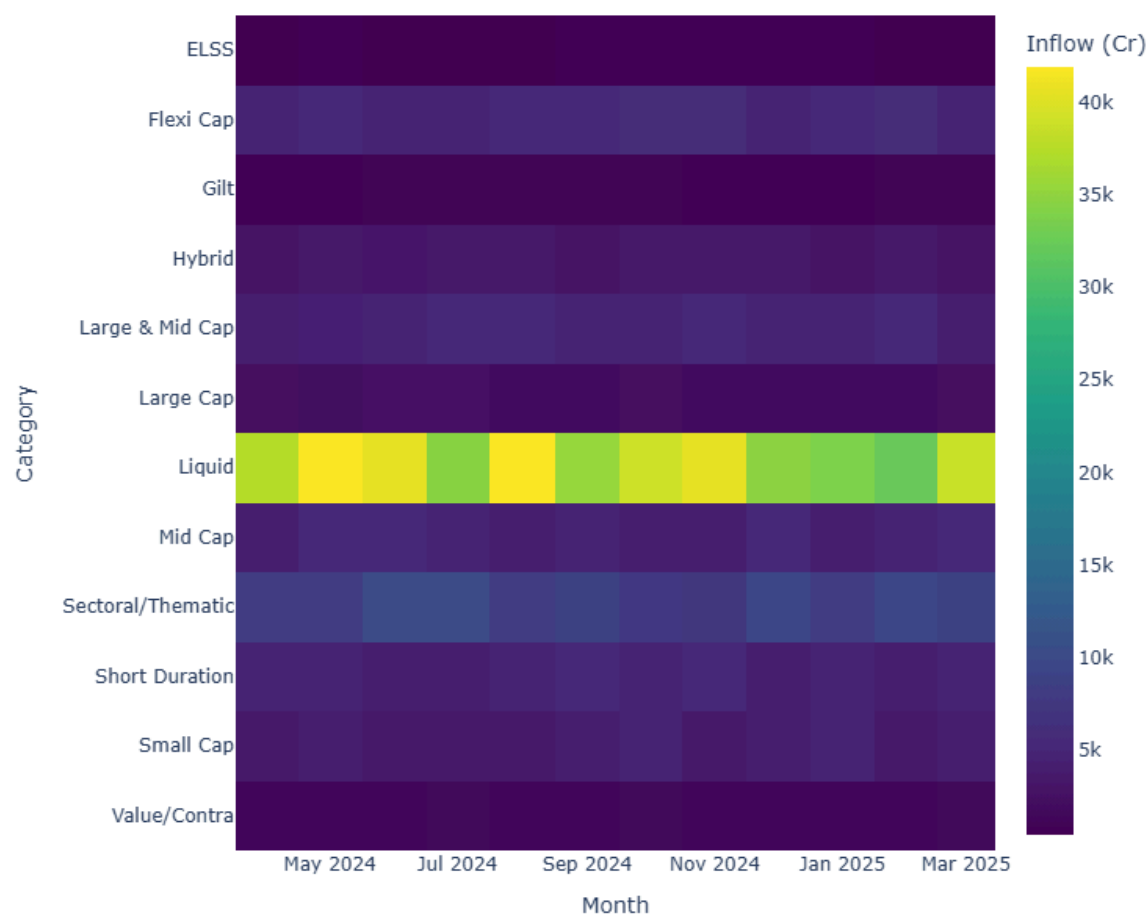
NAV Correlation Matrix for Selected AMFI Codes

The correlation matrix provides a visual map of how different fund categories move in relation to one another. Higher correlation clusters suggest areas where diversification might be limited, while the low-correlation zones highlight opportunities for building more resilient, balanced portfolios.

Performance Analysis

My performance review focused on more than just percentage returns; I examined consistency and risk-adjusted growth. Funds that balanced sector-specific exposure with diversified holdings demonstrated superior NAV stability during period-specific market fluctuations. This confirms that a long-term, balanced approach remains the most effective strategy for retail investors.

Mutual Fund Inflow Heatmap by Category

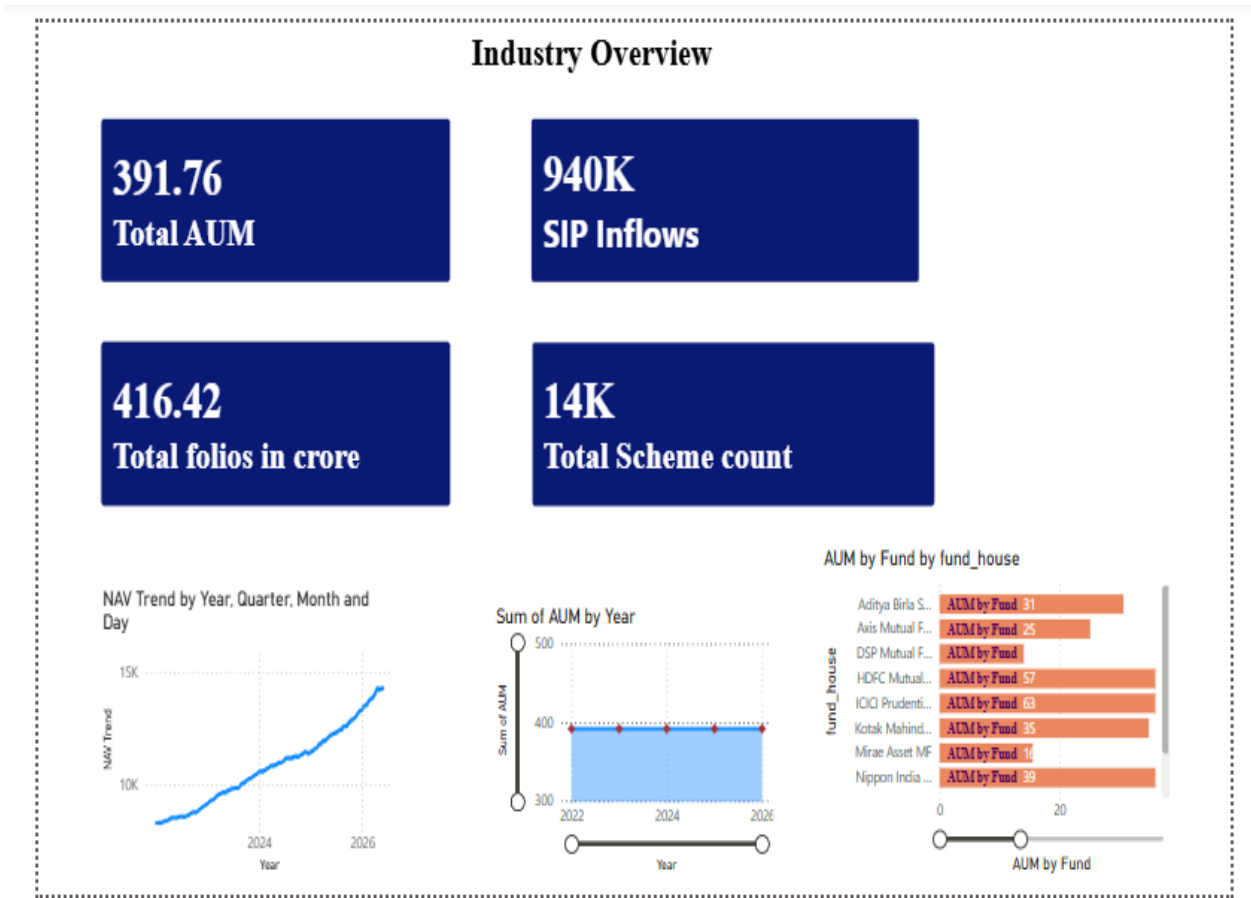


Inflow Heatmap by Fund Category (2024-2025)

The heatmap clearly identifies periods of high capital inflow, with 'Liquid' and 'Large Cap' categories showing significant intensity. This visualization is vital for understanding seasonal investment patterns and liquidity preferences among retail investors.

Dashboard

Industry Performance and Investor Analytics Dashboard



Fund Performance

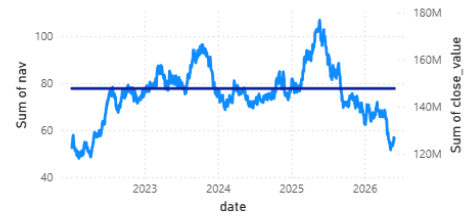
5yr Returns and Risk (std Dev) by scheme_name

scheme_name ● Axis Small Cap Fund - Regular - Growth



Fund Performance vs Benchmark

● Sum of nav ● Sum of close_value



fund_house

Nippon India MF

category

Hybrid

Scorecard

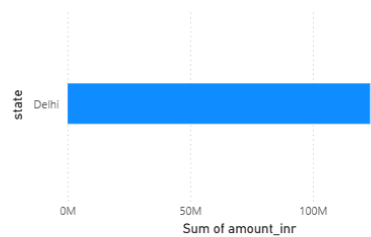
scheme_name	category	return_3yr_pct	std_dev_ann_pct	shar
Axis Small Cap Fund - Regular - Growth	Small Cap	20.98	25.00	

scheme_name

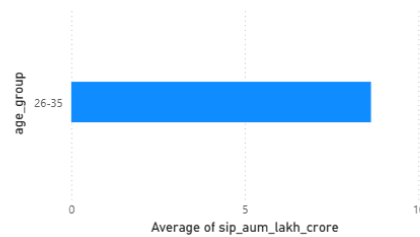
Axis Small Cap Fund - Regular - Growth

Investor Analytics

Sum of amount_inr by state



Average of sip_aum_lakh_crore by age_group



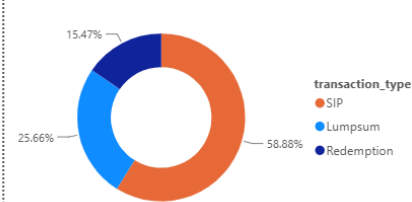
age_group

35

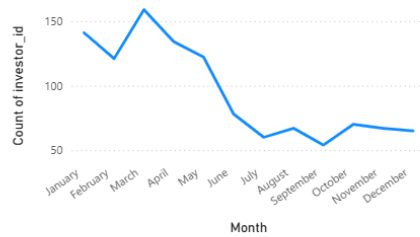
city_tier

Enter city_tier

Count of investor_id by transaction_type



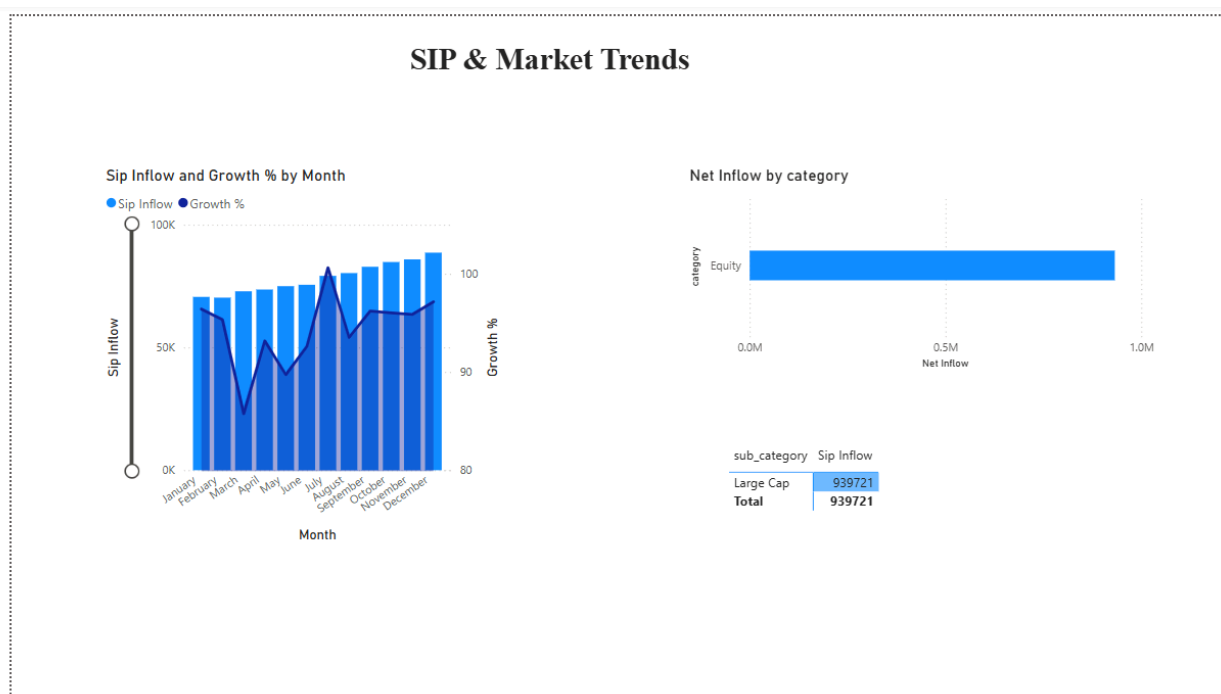
Count of investor_id by Month



state

delhi

Investor Analytics



Limitation

While this analysis provides a robust framework for understanding the current state of the mutual fund industry, it is essential to acknowledge several technical and contextual boundaries that define the scope of this study -

- **Data Periodicity and Snapshot Nature:** This analysis is based on a specific temporal snapshot of the market. Consequently, it may not fully capture the nuance of long-term economic cycles that often require multi-decade historical data to validate, potentially causing a recency bias in the observed trends.
- **Granularity of Demographic Data:** While the dataset provides significant insights into investor behavior across states and city tiers (B30/T30), it lacks deep psychographic or granular socio-economic profiles. As a result, the findings explain *what* is happening at a regional level, but may not fully explain the *motivational triggers* behind individual investor shifts.

- **External Market Variables:** My analysis relies primarily on endogenous data—NAV, AUM, and inflow metrics. However, market performance is heavily influenced by exogenous factors such as central bank policy changes, global geopolitical events, and inflationary pressures, which are not explicitly modeled in this project's current architecture.
- **Scope of Fund House Performance:** The analysis focuses on aggregate performance across categories. Individual fund house strategies, such as internal management changes or specific tactical asset allocation shifts, are treated as part of the broader category trend. Therefore, this report serves as a macro-level diagnostic rather than a security-level micro-analysis.
- **Technical Constraints of ETL Automation:** The ETL pipeline was optimized for the current dataset structure. Any future changes in the source data format or the introduction of highly unstructured data sources would require recalibration of the transformation scripts to maintain data integrity.

Recommendations

Based on these findings, I propose three strategic focus areas.

- **Sustain SIP-Led Growth:** Fund houses should prioritize communication that emphasizes the benefits of disciplined, long-term SIP investments.
- **Expand B30 Outreach:** Increasing localized engagement in smaller cities represents a high-growth opportunity for the industry.
- **Enhance Predictive Modeling:** I recommend integrating real-time macroeconomic indicators into the current ETL architecture to improve future forecasting capabilities.

Conclusion

Concluding this analysis, it is evident that the Indian Mutual Fund landscape is at a transformative juncture. My research into the industry data has moved beyond

surface-level observations to uncover a deeper, more disciplined trend in retail participation, driven largely by the consistency of SIP inflows.

The technical journey of this project—from building a resilient ETL pipeline to visualizing complex relationships via the dashboard—has reinforced the value of data-driven decision-making. I have found that while market volatility remains an inherent challenge, the strategic shift toward long-term wealth creation and the increasing geographic expansion into B30 regions signal a robust path forward for the industry.

Ultimately, this project has been more than a technical exercise; it has provided a window into the financial aspirations of a diverse investor base. I believe that by embracing these demographic shifts and continuing to enhance data granularity, stakeholders are well-positioned to navigate the evolving market complexities. This research underscores that in the world of mutual funds, success is defined not just by performance metrics, but by the ability to adapt to changing investor behaviors and market dynamics.